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Protocol modified from Carolina BioKits

Protocol: Looking at mitosis using an onion root tip

Working in pairs, you will perform the following experiment and collect data.

1) Using forceps, remove your onion root for the Carnoy's solution placed inside the fume hood and put into your plastic cup to transport back to your workstation.

2) Place your onion root onto a microscope slide.

3) Using the provided razor blade, cut off the opposite end of the root tip and leave 1-2mm of root tip on your slide. The remaining part of the root can be placed on a paper towel and disposed of.

4) Add 1-2 drops of Toluidine blue O (TBO) dye to your root tip. Make sure it is fully submerged. Leave submerged for 2 minutes.

5) Remove the dye using the corner of a Kimwipe or paper towel. Make sure NOT to touch the onion root tip with your towel or it may stick and you may lose it.

6) Add 1-2 drops of water from the dropper bottle to your root tip and place a cover slip over top.

7) Using a Kimwipe, place over the cover slip and apply even pressure with your thumb to press straight down on the root tip to crush and flatten the cells. After doing, check to see if the root tip is flattened (if needed, you can repeat this to flatten more).

8) Place your slide under the compound microscope and, using the 4x objective lens, bring your slide into focus. Once you can see individual cells, you can increase the magnification to moving to the 10x and then 40x objective lenses.

9) **Draw this table on your notebook to calculate the mitosis index**

Stage of mitosis	Tally (observed # of cells)	Percentage (%)
Interphase		
Prophase		
Metaphase		
Anaphase		
Telophase		
TOTAL	**50**	**100%**